6.3.1 Chromatography and qualitative analysis

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
Types of chromatography		
(a)	interpretation of one-way TLC chromatograms in terms of R_f values	*M3.1* **PAG6** HSW3 Interpretation of TLC to analyse organic compounds.
(b)	interpretation of gas chromatograms in terms of: **(i)** retention times (ii) the amounts and proportions of the components in a mixture.	*M3.1, M3.2* To include creation and use of external calibration curves to confirm concentrations of components. Peak integration values will be supplied. HSW3 Interpretation of GC to analyse organic compounds.
Tests for organic functional groups		
(c)	qualitative analysis of organic functional groups on a test-tube scale; processes and techniques needed to identify the following functional groups in an unknown compound: **(i)** alkenes by reaction with bromine **(see also 4.1.3 f)** **(ii)** haloalkanes by reaction with aqueous silver nitrate in ethanol **(see also 4.2.2 a)** (iii) phenols by weak acidity but no reaction with CO_3^{2-} **(see also 6.1.1 h)** (iv) carbonyl compounds by reaction with 2,4-DNP **(see also 6.1.2 d)** (v) aldehydes by reaction with Tollens' reagent **(see also 6.1.2 e)**	**PAG7** HSW4 Qualitative analysis.

(vi) primary and secondary alcohols and aldehydes by reaction with acidified dichromate **(see also 4.2.1 c, 6.1.2a)**

(vii) carboxylic acids by reaction with CO_3^{2-} **(see also 6.1.3 b)**.